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BE (6th Semester)

Examination, Nov.-Dec., 2014

Branch : Electrical, EEE

POWER ELECTRONICS

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : Part (a) is compulsory. Attempt any two parts out of (b), (c) & (d).

Q. 1. (a) What is reverse recovery current of an SCR. 2

(b) Derive an expression for anode current of a thyristor from two transistor analogy. 7

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(c) With neat sketches of the turn on and turn off characteristics of SCR

(d) A relaxation oscillator using an UJT, is to be designed for triggering an SCR. The UJT has the following data : 7

$\eta = 0.72$, $I_p = 0.6$ mA, $V_p = 18.0$ V, $V_v = 1.0$ V

$I_v = 2.5$ mA, $R_{BB} = 5$ KΩ

Normal leakage current with emitter open = 4.2 mA. The firing frequency is 2 kHz for C = 0.04 μf, compute the value of R, R_1 and R_2.

Q. 2. (a) State the difference between half wave controlled and half controlled converter. 2

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(b) Draw neat sketches of the output voltage & current waveform of a single phase fully controlled bridge converter. 7

(c) Compare the symmetrical and asymmetrical configuration of single phase half controlled rectifier. 7

(d) A 230 V, 50 Hz, one-pulse SCR controlled converter is triggered at a firing angle of 40° and the load current extinguishes at an angle of 210°. Find the circuit turn off time, average output voltage and the average load current for : 7

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R = 5 Ω and L = 2 mH

R = 5 Ω and L = 2 mH and E = 110 V

Q.5 a How can a conducting thyristor be turned off ?

b Draw the circuit of Morgan chopper and explain its working. Draw waveforms of different current and voltages 7

c Explain with circuit diagram the four quadrant operation of class - E chopper 7

d A voltage commutated copper has the following parameters : V_s = 220 V, load circuit parameters are 0.5 Ω, 2 mH, 10 V, commutation circuits parameters are L = Q.

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20 μH, C = 50 μF, T_{on} = 800 μ sec, T = 2000 μ sec. For a constant load current of 80 amp. Compute the following : 7

(i) Effective on period

(ii) Peak current through main thyristor T_1 & auxillary thyristor T_A.

(iii) Turn off times for T_1 & T_A.

(iv) Total communication interval.

Q. 4. (a) Write any two application of cyclo converters. 2

(b) Explain the operation of single phase to single phase cyclo converter feeding R and R - L load. Draw the appropriate waveforms. 7

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(c) State two limitation of basic series inverter & how it is overcome by improved series resonant inverter. 7

(d) Explain the working of individual commutation in single phase full bridge McMurray inverter with relevant waveforms. 7

Q. 5. (a) List two advantages of power MOSFET over power BJT. 2

(b) Explain IGBT as per the following points : 7

(i) Forward blocking capacity

(ii) Reverse blocking capacity

(iii) Forward conduction mode

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(c) A DC shunt motor operating from a single phase half controlled bridge converter runs at 1450 rpm and has a back emf of 75 V. The supply voltage is 230 V rms, 50 Hz. The SCR's are fired symmetrically of firing angle of $\pi/4$ rad in each half cycle. Neglect armature inductance find the conduction angle and average voltage applied to the armature. 7

(d) A chopper used for on-off control of a DC shunt motor has a supply voltage 220 V. On time of 10 m sec. and off time of 12 m sec. the armature resistance is 3 Ω. Neglecting

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armature inductance and assuming continuous conduction of motor current. Calculate the average load current when the motor runs at 1400 rpm. And has a voltage constent K_v = 0.495. (K_v = Back emf / motor speed in rad/ sec) 7

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